Measuring the Radius of a Star (10 points)

Measuring the radii of stars is difficult because of their huge distances. Inhomogeneities of the atmosphere blur the image and make it hard to determine their actual size. One possible method is to use an interferometer, proposed by Michelson and Pease in 1921 to measure the diameter of Betelgeuse. The figure below shows the interferometer used to measure the radius of a star.

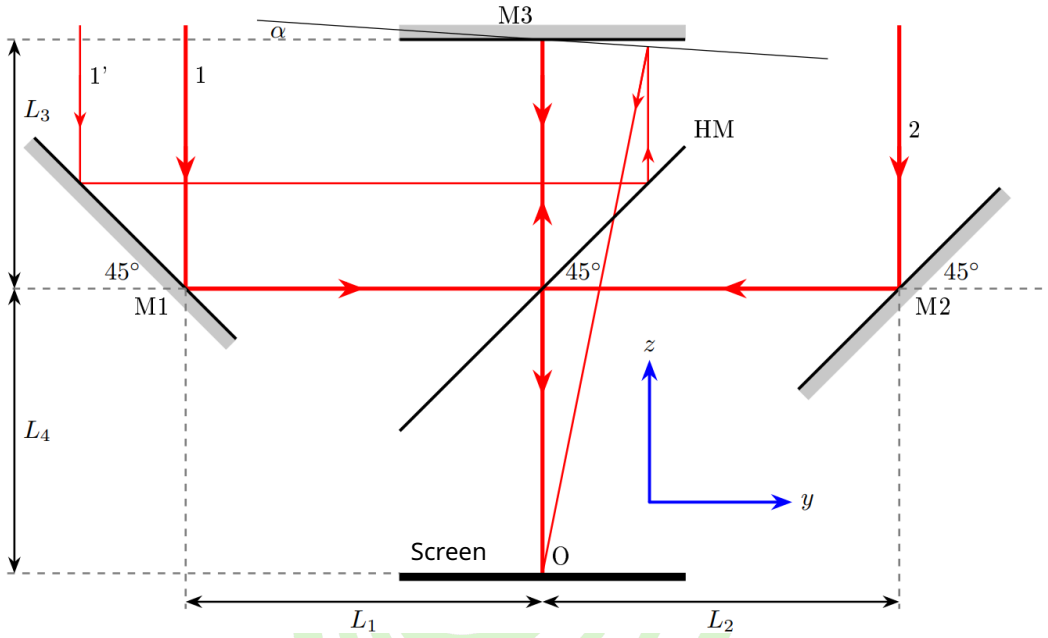


Fig. 1. Paths of the light rays coming from the star.

Reflected by mirror M1, ray 1 reaches the beam splitter HM, which reflects 50% of the light intensity and transmits the other 50%. The reflected ray goes to mirror M3, is reflected, passes through HM and falls on the screen. Meanwhile ray 2 is reflected by mirror M2; reflected by HM, it also reaches the screen. Mirrors M1, M2 and M3 reflect all the light falling on them and add π radians to its phase. The wavelength of the light is λ.

Part A. Rays parallel to the z axis (1.6 points)

A.1	Consider the case where rays 1 and 2 are parallel to the z axis and mirror M3 is parallel to the xy plane. Find the optical path difference Δ_{12} between rays 1 and 2 if both reach the screen at point O.	0.2 pt.

A.2	Now mirror M3 is tilted by an angle α from the xy plane by a rotation about the x axis (see Fig. 1). If ray 1′ reaches the screen at point O, find the optical path difference $\Delta_{1'2}$ between rays 1′ and 2. Show that to first order the path difference does not depend on α, and simplify your answer for $\alpha \ll 1$.	1.4 pt.

Part B. Rays at an angle θ in the yz plane (4.7 points)

Now the rays enter the interferometer at an angle $\theta \ll 1$ to the z axis in the yz plane, as shown below. As in A2, mirror M3 is tilted by α.

Note: from here on, work to first order when computing optical paths, i.e. neglect terms containing α^2, θ^2, $\alpha\theta$ and so on.

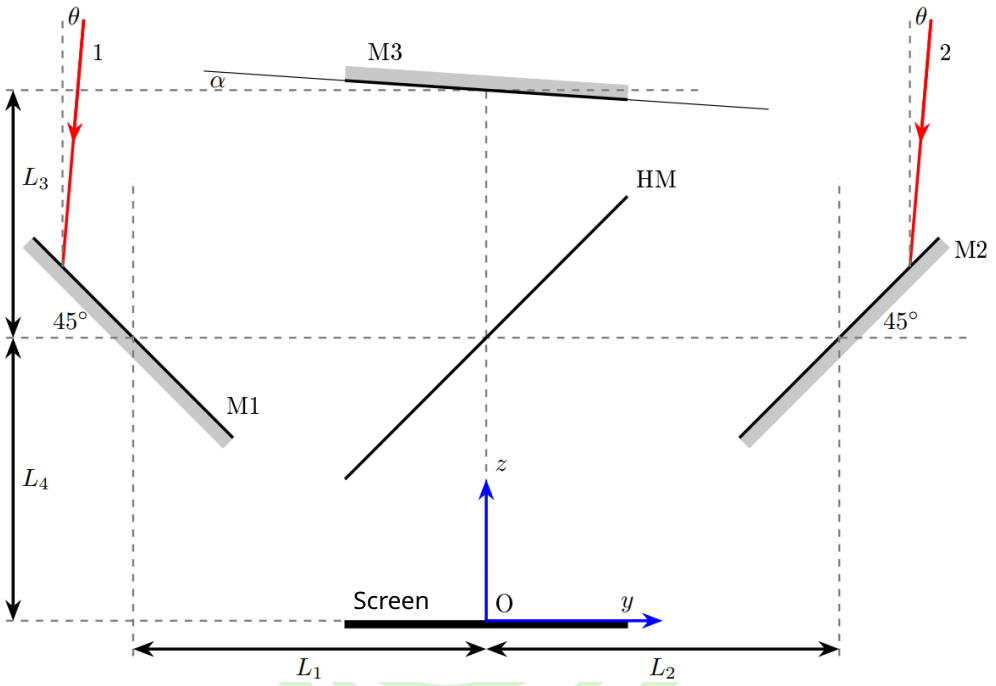


Fig. 2. Rays travelling at an angle θ in the yz plane and reaching the screen at point O.

B.1	Let rays 1 and 2 hit fixed points of mirrors M1 and M2 respectively. Show on the working sheets that, to first order, the optical paths of rays 1 and 2 from their points of reflection on M1 and M2 to the screen do not depend on the angles θ and α.	0.8 pt.

B.2	Let the angles α and θ be fixed while the points of incidence of rays 1 and 2 on M1 and M2 may change. Show on the working sheets that the optical paths of rays 1 and 2 from the dashed line at $z_\mathrm{d} = L_3 + L_4$ (its intersections with the paths of rays 1 and 2 before they hit M1 and M2) to the screen do not depend, to first order, on θ and α.	0.8 pt.

B.3	If rays 1 and 2 reach the screen at point O, their phase difference can be written as $$\delta_{12} = \frac{2\pi}{\lambda} b\theta + \delta_O.$$ Express b and δ_O in terms of λ, L_1, L_2, L_3, L_4.	1.0 pt.

B.4	If ray 1′ reflected by M1 and ray 2′ reflected by M2 reach the same point of the screen at coordinate y, their phase difference can be written as $$\delta_{1'2'} = \frac{2\pi}{\lambda} b\theta + \delta_y.$$ Show on the working sheets that $$\delta_y = \delta_O - \frac{2\pi}{\lambda} \cdot 2\alpha y.$$ *Note:* in all the following tasks you may use the statements of B1–B4 even if you have not proved them.	0.6 pt.

B.5	An interference pattern thus appears on the screen. Find the distance y_1 from O to the nearest interference maximum, and the distance Δy_m between two neighbouring maxima. Give numerical answers for $L_1 = 0.50$ m, $L_2 = 1.50$ m, $L_3 = 0.50$ m, $L_4 = 1.00$ m, $\alpha = 10^{-4}$, $\theta = 10^{-5}$, $\lambda = 500$ nm.	0.6 pt.

B.6	Find how the intensity $I(y)$ on the screen depends on y if its maximum value is I_0, substituting the numerical values of the previous task. Take into account the effect of the beam splitter on the intensity of the transmitted light.	0.9 pt.

Part C. Quite arbitrary rays (1.0 point)

Now rays 1″ and 2″ enter the interferometer at an angle $\theta_x \ll 1$ to the yz plane and an angle $\theta_y \ll 1$ to the xz plane (see Fig. 3).

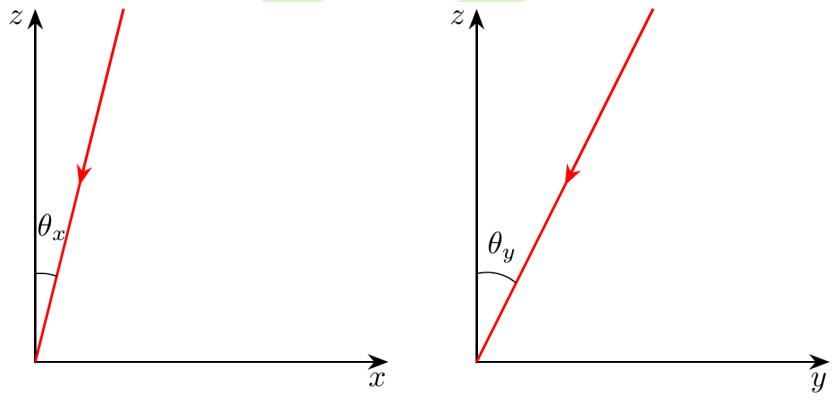


Fig. 3.

C.1	Express the wave vector $\vec{k}$ of the incident rays in terms of θ_x, θ_y, the magnitude $k = 2\pi/\lambda$ and the unit vectors of the axes.	0.4 pt.

C.2	Let ray 1″ reflected by M1 and ray 2″ reflected by M2 reach the same point of the screen on the y axis, a distance y from O. Express the phase difference between rays 1″ and 2″ in terms of λ, L_1, L_2, L_3, L_4, θ_x, θ_y, α, y.	0.6 pt.

Part D. Measuring the radius (2.7 points)

Consider a star of angular radius θ_R, which we want to determine. For simplicity the star radiates as a uniform disc: the intensity of the light coming from a solid angle $\mathrm{d}\Omega = \mathrm{d}\theta_x \mathrm{d}\theta_y$ is $\mathcal{J}\mathrm{d}\Omega$, with $\mathcal{J}$ the same in every direction within the stellar disc. The interferometer (its z axis) points exactly at the centre of the star. For $\alpha > 0$ an interference pattern is seen on the screen. Let $I_{y(\min)}$ and $I_{y(\max)}$ be the minimum and maximum intensities on the screen along the y axis, and define the visibility

$$V = \frac{I_{y(\max)} - I_{y(\min)}}{I_{y(\max)} + I_{y(\min)}}.$$

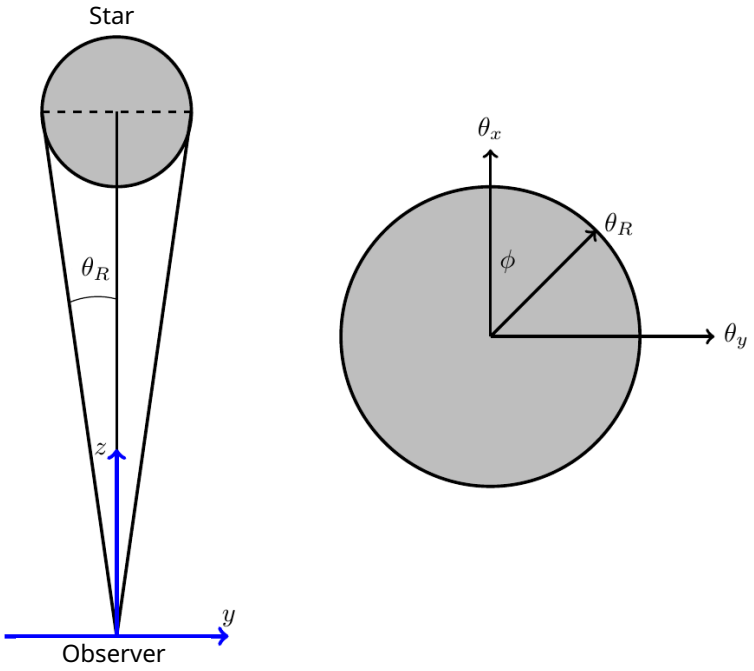


Fig. 4. Left: side view of the star and the observer; right: the stellar disc seen by the observer.

D.1	Write V as an integral over one variable. You need not carry out the integration. *Hint:* the integral of a function $f(\theta_y)$ of θ_y only over a disc of radius θ_R can be written as $$I = \int_{-\theta_R}^{\theta_R} \mathrm{d}\theta_y \int_{-\sqrt{\theta_R^2-\theta_y^2}}^{\sqrt{\theta_R^2-\theta_y^2}} \mathrm{d}\theta_x \, f(\theta_y) = \int_{-\theta_R}^{\theta_R} \mathrm{d}\theta_y \cdot 2\sqrt{\theta_R^2 - \theta_y^2}\, f(\theta_y).$$	1.8 pt.

D.2	The interferometer is used to observe Betelgeuse. If mirrors M1 and M2 are moved symmetrically away from the centre of HM (so that $L_1^* = L_1 + d/2$ and $L_2^* = L_2 + d/2$), the visibility V vanishes at $d = 0.97$ m, 3.12 m, ... (the other parameters as in B3; here L_1 and L_2 are not given numerically, $\lambda = 500$ nm). Find the angular radius θ_R of Betelgeuse, and its radius R if its distance is $h = 6.079 \cdot 10^{15}$ km. *Hint:* the first two zeros of $F_n = \int_0^1 (1-w^2)^n \cos(\eta w)\,\mathrm{d}w$ as a function of η are: n | η_1 | η_2 0.5 | 3.832 | 7.016 0 | 4.493 | 7.725 1.5 | 5.136 | 8.417	0.9 pt.

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